

Formulas*discrete time growth:*

- $N_T = N_0 \lambda^T$
- $\lambda = f + p$
- $\mathcal{R} = f/(1 - p)$

continuous time growth:

- $N(t) = N(0) \exp(rt)$
- $r = b - d$
- $\mathcal{R} = b/d$

structured growth:

- $\ell_x = p_1 \times p_2 \times \dots p_{x-1}$
- $\mathcal{R} = \sum \ell_x f_x$

competition:

- α_{ij} = effect **of** species i **on** species j
- $C = \alpha_{12}\alpha_{21}$
- $E_{ij} = \alpha_{ij}K_i/K_j$

Use the following information for the next two questions. A population of hemlock trees is estimated to be at stable age distribution, with a constant life table, with reproductive number $\mathcal{R}= 1.2$. It takes the trees several decades to reach maturity and reproduce.

1. This population is

- A. declining
- B. stable
- C. increasing
- D. showing damped oscillations
- E. there is not enough information to answer this question

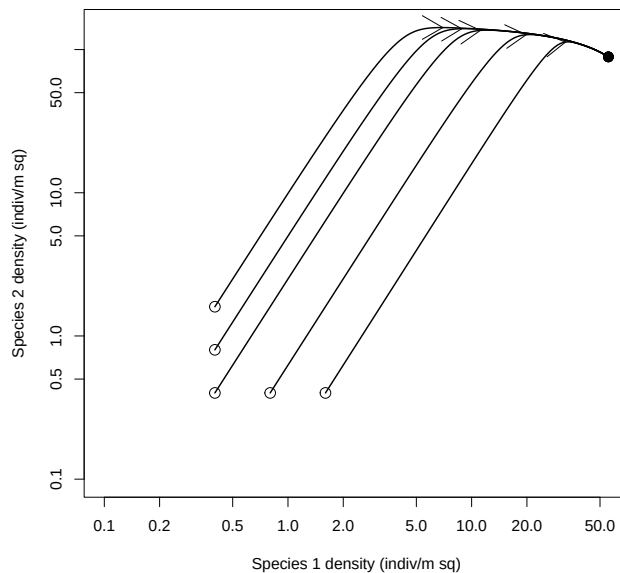
2. What is the *most accurate* statement you can make about the finite rate of growth λ , measured with a time step of one year?

- A. We expect $\lambda > 1.2$
- B. We expect $\lambda = 1.2$
- C. We expect $\lambda > 1$
- D. We expect $1.2 > \lambda > 1$
- E. We expect $1.2 > \lambda$

3. We expect populations to be regulated (kept under control). This typically works because their _____ tends to decrease when the population _____.

- A. growth rate; has been established for a long time
- B. growth rate; becomes larger
- C. projected future size; has been established for a long time
- D. projected future size; becomes larger

Use the picture above for the next three questions. The path in the middle corresponds to a starting density of 0.4 indiv/m² for each species.



4. This picture shows what sort of competition outcome?

- A. Species 1 dominates
- B. Species 2 dominates
- C. The species co-exist
- D. There is mutual exclusion

5. Assuming density dependence can be neglected at low densities, which species has the higher value of $r_{\max}$?

- A. Species 1
- B. Species 2
- C. They have the same value of $r_{\max}$
- D. Either species can have a higher value, depending on initial conditions.

6. From this picture, we can conclude that:

- A. Both of the competition coefficients $\alpha < 1$.
- B. Both of the competition coefficients $\alpha > 1$.
- C. Both of the population-level competitive effects $E < 1$.
- D. Both of the population-level competitive effects $E > 1$.
- E. We cannot make any of these conclusions from this picture.

7. A simple population model has *structure*, but not *regulation* (individuals are assumed to be independent). Cohorts are not independent. If the model has $\mathcal{R}_0 > 1$, then: The modeled population _____ grow exponentially at first, and _____ grow exponentially as it approaches a stable age distribution (SAD)

- A. will; will
- B. may not; will
- C. will; may not
- D. may not; may not

8. We believe that in a particular desert, Joshua trees and cactuses are competing only for water. According to our simple theory, we would expect:

- A. A tendency for co-existence
- B. The species with the larger value of K (indiv/ha) to dominate
- C. The species with the larger value of α (measuring its impact on the other species) to dominate
- D. The species with the larger value of E (measuring its impact on the other species) to dominate
- E. A tendency for mutual exclusion

9. Japanese maples are trying to invade a stable Ontario forest dominated by sugar maples (we don't know whether they will be successful). Based on these assumptions, we would expect the reproductive number of the _____ population to satisfy _____.

- A. sugar maple; $\mathcal{R} > 1$
- B. Japanese maple; $\mathcal{R} > 1$
- C. sugar maple; $\mathcal{R} = 1$
- D. Japanese maple; $\mathcal{R} = 1$

10. Which of these is *not* an effect that tends to *balance* the tendency of populations to grow or decline exponentially without limit?

- A. Allee effects
- B. Competition for access to mates
- C. Competition for access to resources
- D. Resource depletion
- E. Dynamics of natural enemies

11. A scientist has made a life table for a perennial population of deer, with censusing *before* reproduction. If she re-does the life table with a census after reproduction, she would expect $\mathcal{R}$ to be _____ because population dynamics look _____ when viewed before reproduction.

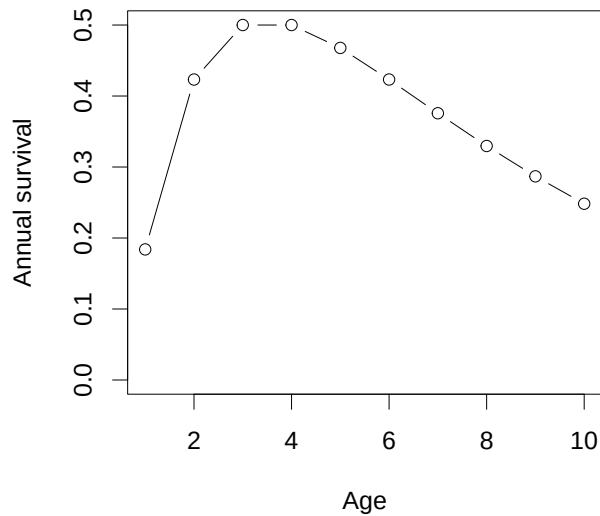
- A. higher; fastest
- B. higher; slowest
- C. lower; fastest
- D. lower; slowest
- E. the same; the same

12. The “balance argument” for sexual allocation implies that, if individuals do not interact strongly with their siblings, females should on average:

- A. Produce the same number of male and female offspring
- B. Use the same amount of resources *per offspring* for male and female offspring
- C. Use the same *total* amount of resources for male and female offspring
- D. All of the above

13. A species of small plant is highly adapted to dramatic changes in conditions (ranging from very good growth conditions to very bad). We would expect it to:

- A. Have a very fast life cycle
- B. Have a very slow life cycle
- C. Have adaptations that speed up its life cycle in good conditions and slow it down in bad conditions
- D. Have adaptations that speed up its life cycle in bad conditions and slow it down in good conditions

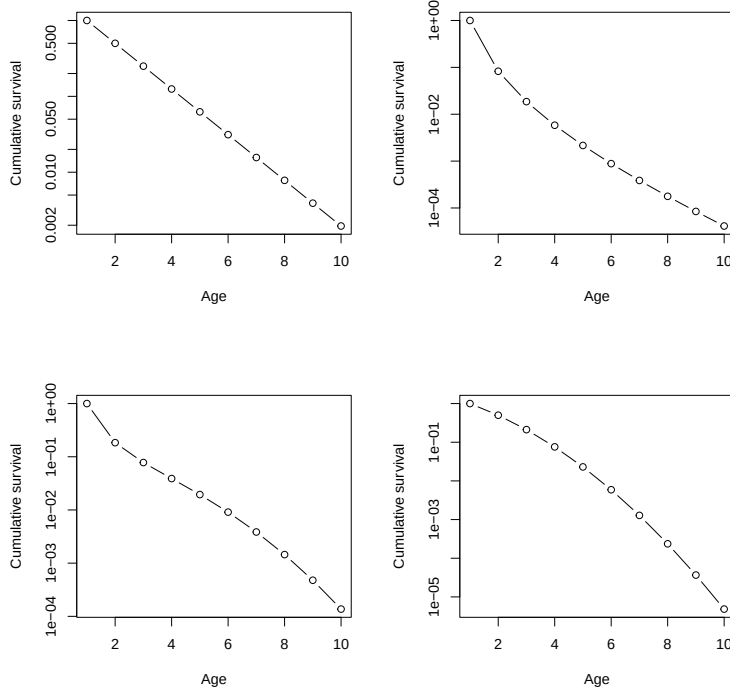


Use the picture above for the following 2 questions.

14. What does this picture of survivorship in an idealized age-structured population indicate about *mortality* in this population?

- A. Mortality is constant
- B. Mortality is elevated in older individuals
- C. Mortality is elevated in younger individuals
- D. Mortality is elevated in both older and younger individuals

15. The pictures below show *cumulative* survival. Which one corresponds to the picture shown above?

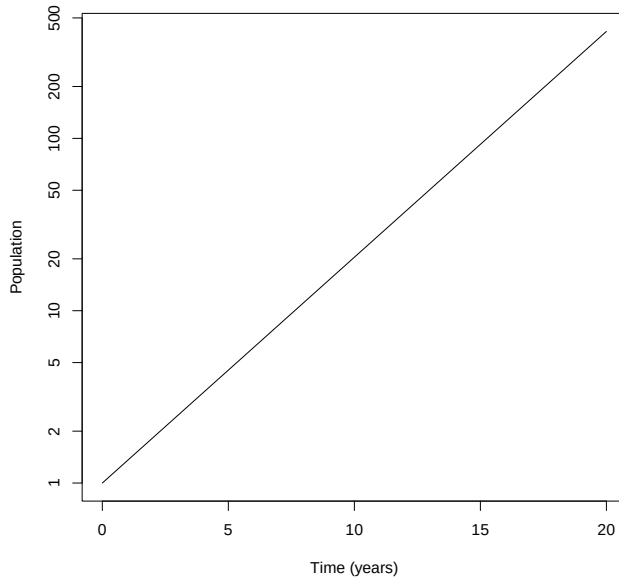


- A. Upper left
- B. Upper right
- C. Lower left
- D. Lower right

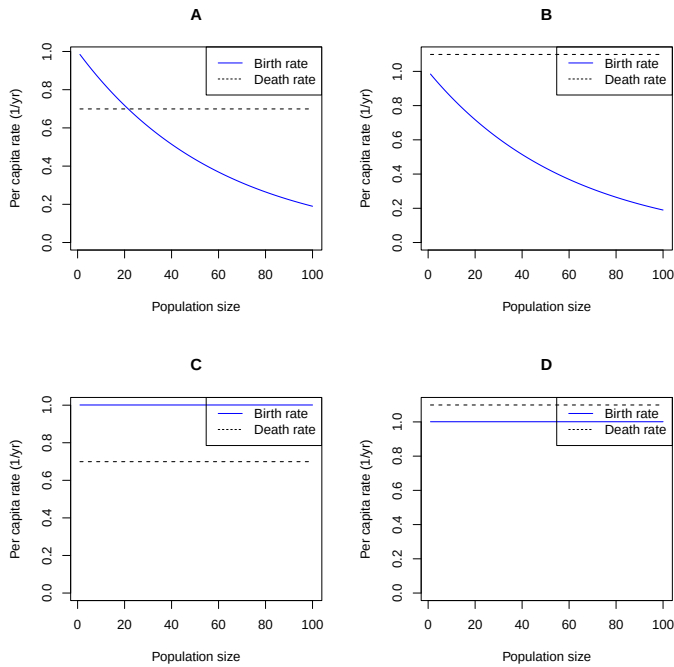
16. Which of the following is *not* a possible advantage of producing fewer offspring with the same total amount of energy (thus investing more in each offspring)?

- A. Greater ability to help offspring survive
- B. Greater ability to help offspring disperse
- C. Greater ability to produce high-quality offspring
- D. More energy left over to produce offspring in future years

Use the picture below for the next two questions. It shows a time series for a continuous-time birth-death model.



17. This picture shows a population that is:
- A. Increasing arithmetically
 - B. Increasing geometrically
 - C. Increasing arithmetically on the log scale, but geometrically on a linear scale
 - D. Increasing geometrically on the log scale, but arithmetically on a linear scale
18. Which of the small pictures shows the assumptions that generated the time plot?



- A. A
- B. B
- C. C
- D. D

19. In an environment where seed survival is highly variable from year to year, we expect that semelparous plants (reproducing only once) are likely to invest relatively _____ than iteroparous plants in _____.

- A. more; mechanisms for seed dispersal
- B. less; mechanisms for seed dispersal
- C. more; each individual seed
- D. less; each individual seed

20. Which of the following traits is *not* typically associated with r strategies?

- A. Fast life cycle
- B. Efficient resource use
- C. Relatively small investment per offspring
- D. Relatively large investment in dispersal

Short-answer questions

Answer questions *in pen*. *Briefly* show necessary work and equations. Points may be *deducted* for wrong information, even when the correct information is also there.

21. An annual plant has variable fecundity: each female has an average of 3 successful female offspring in an good year (half of the time) and an average of 0.6 successful female offspring in a bad year (the other half of the time).

a) (1 point) What is the long-run average λ for this species?

b) (1 point) What are two possible strategies this plant could evolve to spread out the risk of a bad year (hedge its bets)?

22. A species of falcon and a species of hawk are living in the same region and competing to eat songbirds and small mammals that they find there.

a) What is a possible reason why the competition between species might be relatively *stronger* than within species in this case?

b) What is a possible reason why the competition between species might be relatively *weaker* than within species in this case?

c) What are *two* possible outcomes of the competitive interaction in the second case (when between-species competition is weak)?